CHL5233: Exploratory Data Analysis

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First look at the data

- variable class and sample size

Variable	Class
Country (ISO code)	character
Year	integer
Maternal mortality	numeric
Infant mortality	numeric
Neonatal mortality	numeric
Under-5 mortality	numeric
GDP per capita	numeric
OECD member	factor
Population density	numeric
Urban residence	numeric
Age dependency ratio	numeric
Male education	numeric
Temperature	numeric
Rainfall	numeric
Drought	factor
Earthquake	factor
Battle-related deaths	numeric
Armed conflict	factor
Conflict intensity	ordered
Battle-related deaths: none or positive-death quintile	ordered

Country (ISO code)	Year	Maternal mortality	Infant mortality	Neonatal mortality	Under-5 mortality	GDP per capita	Rainfall	Male education	Armed conflict
OMN	2006	18	10.7	5.6	12.5	14.5337294	0.1255374	6.400321	0
KHM	2003	404	61.8	29.7	78.0	0.3642753	1.6418610	4.272785	0
ZMB	2018	NA	43.9	24.7	65.4	1.4752045	1.0730969	8.756643	0
IND	2011	197	43.0	30.6	55.0	1.4496018	1.2775913	6.606593	1
BLR	2019	NA	2.4	1.1	3.1	6.8377178	0.7443459	12.229370	0
MRT	2017	766	52.6	33.1	77.7	1.6346436	0.2151420	6.691080	0
RWA	2002	961	90.3	37.3	149.6	0.2346824	1.1770086	3.695169	1
SEN	2000	553	67.7	38.2	129.5	0.6196421	0.6895503	2.375009	1
BTN	2007	275	40.1	25.3	50.8	1.7140325	2.3876547	2.517513	0
MWI	2013	396	42.2	24.8	63.4	0.3443968	1.0692260	6.527678	0

Individual variables

The table below gives the type, observed range, and overall missingness of each analysis variable.

Variable	Class	Missing (n)	Missing (%)	Minimum	Median	Maximum
Maternal mortality	numeric	426	0.115	2.000	66.000	2480.000
GDP per capita	numeric	62	0.017	0.110	4.072	123.679
Infant mortality	numeric	20	0.005	1.600	18.900	138.100
Male education	numeric	20	0.005	1.067	8.368	14.441
Neonatal mortality	numeric	20	0.005	0.800	12.100	60.900
Population density	numeric	20	0.005	0.000	27.519	99.860
Rainfall	numeric	20	0.005	0.020	1.013	4.711
Temperature	numeric	20	0.005	-2.405	21.958	29.676
Under-5 mortality	numeric	20	0.005	2.000	22.200	224.900
Urban residence	numeric	20	0.005	0.103	30.254	93.413
Age dependency ratio	numeric	0	0.000	16.172	55.512	111.477
Armed conflict	factor	0	0.000	NA	NA	NA
Battle-related deaths	numeric	0	0.000	0.000	0.000	78644.000
Battle-related deaths: none or positive-death quintile	ordered	0	0.000	NA	NA	NA
Conflict intensity	ordered	0	0.000	NA	NA	NA
Drought	factor	0	0.000	NA	NA	NA
Earthquake	factor	0	0.000	NA	NA	NA
OECD member	factor	0	0.000	NA	NA	NA

visualizing year pattern in missing maternal mortality data

```
# A tibble: 38 x 3
# Groups:   Missing Maternal Mortality [2]
   `Missing Maternal Mortality`  year     n
   <lgl>                        <int> <int>
 1 FALSE                         2000   183
 2 FALSE                         2001   183
 3 FALSE                         2002   183
 4 FALSE                         2003   183
 5 FALSE                         2004   183
 6 FALSE                         2005   183
 7 FALSE                         2006   183
 8 FALSE                         2007   183
 9 FALSE                         2008   183
10 FALSE                         2009   183
# i 28 more rows
```

There's >1% missingness in two variables. Maternal mortality is missing in 11.5% of observations, as it is only observed from 2000-2017. GPD per capita is missing in 1.7% of observations.

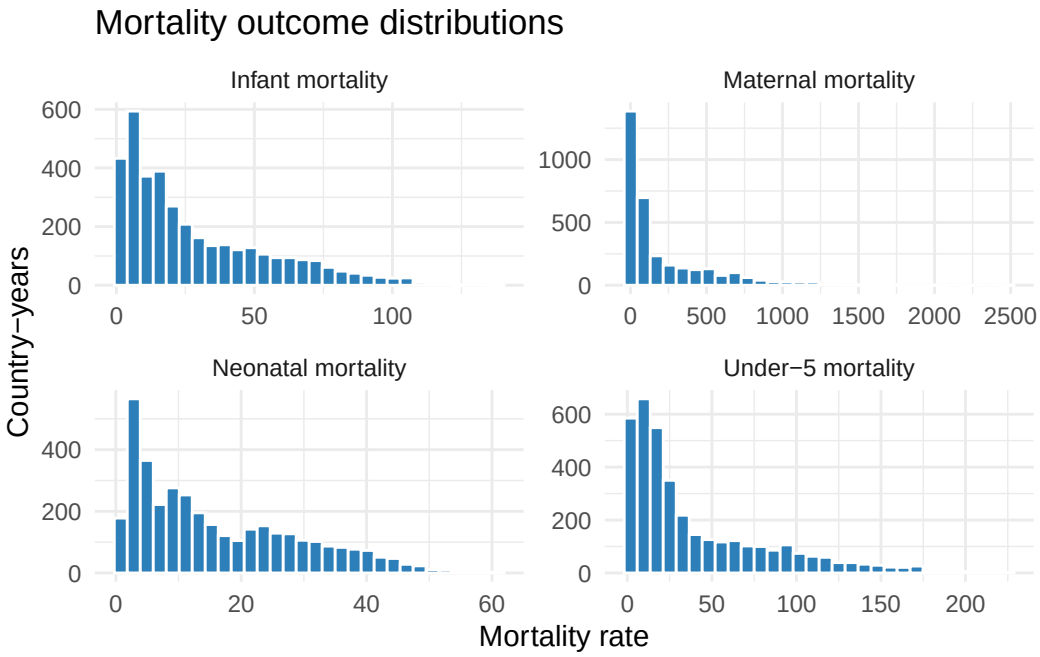


Figure 1: Distributions of mortality outcomes across country-years.

Missing data

The following plots show whether missingness is patterned by time, country, or conflict status.

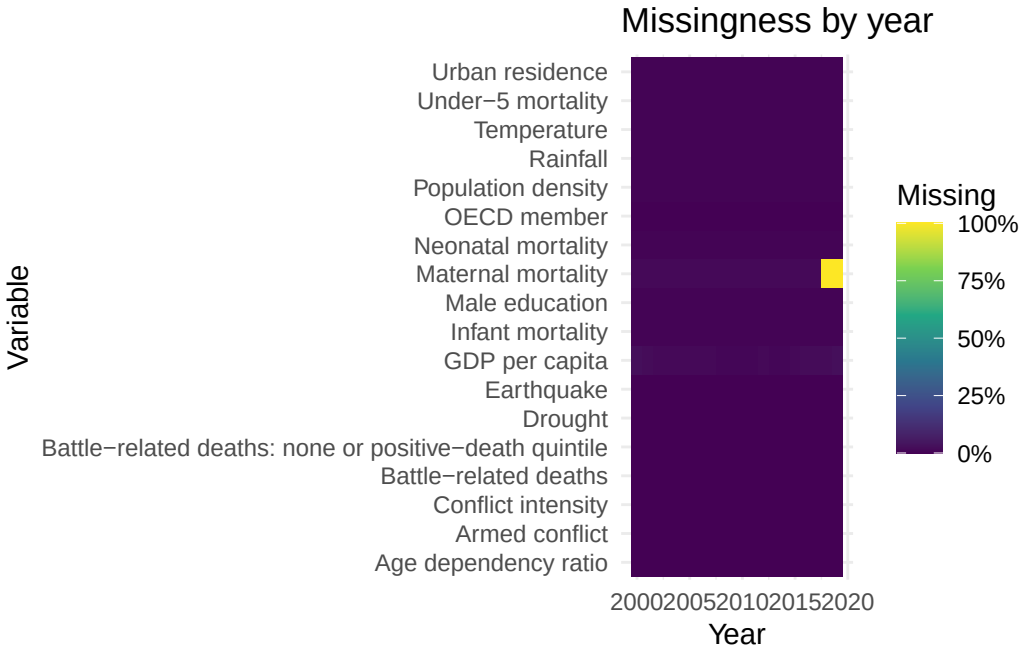


Figure 2: Proportion missing for each variable in each year.

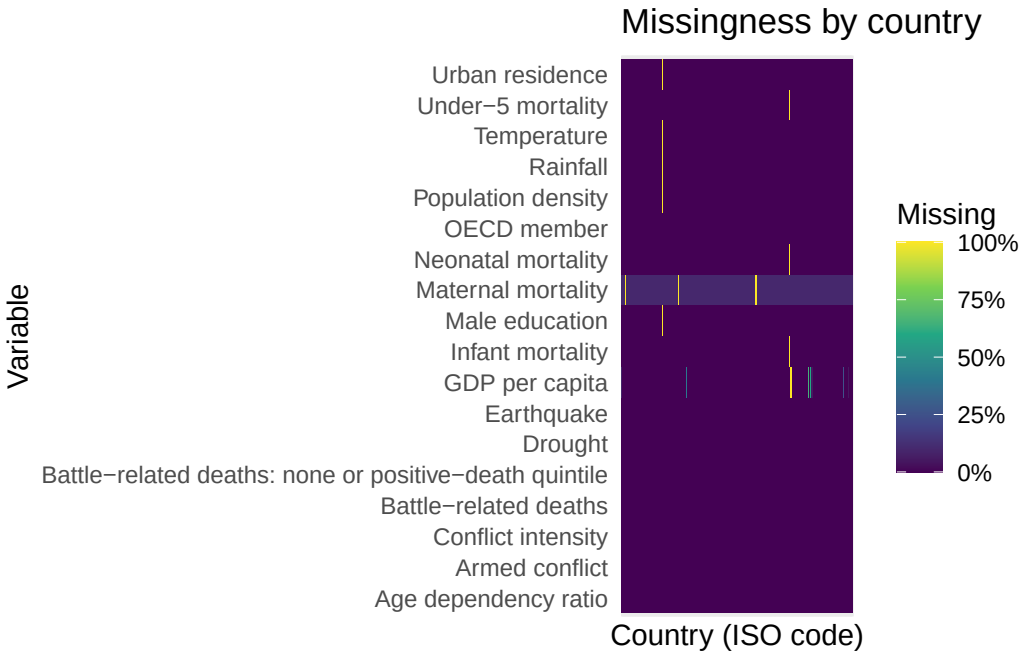


Figure 3: Proportion missing for each variable within each country over 2000–2019.

viewing some data where variables are missing

Country (ISO code)	Year	Maternal mortality	Infant mortality	Neonatal mortality	Under-5 mortality	GDP per capita	Rainfall	Male education	Armed conflict
PRI	2000	26	NA	NA	NA	16.19213	1.547357	10.83336	0
PRI	2001	26	NA	NA	NA	18.12320	1.861477	10.91577	0
PRI	2002	24	NA	NA	NA	18.73146	1.747010	10.99659	0
PRI	2003	23	NA	NA	NA	19.55712	2.369716	11.07591	0
PRI	2004	25	NA	NA	NA	20.98899	2.497150	11.15385	0
PRI	2005	23	NA	NA	NA	21.95932	2.650943	11.23048	0
PRI	2006	23	NA	NA	NA	22.93594	2.630250	11.30594	0
PRI	2007	23	NA	NA	NA	23.66488	2.112395	11.38032	0
PRI	2008	23	NA	NA	NA	24.89833	2.329006	11.45377	0
PRI	2009	23	NA	NA	NA	25.76873	2.385495	11.52640	0
PRI	2010	21	NA	NA	NA	26.43575	3.066733	11.59832	0
PRI	2011	22	NA	NA	NA	27.27888	2.935664	11.66964	0
PRI	2012	19	NA	NA	NA	27.94473	1.878001	11.74046	0
PRI	2013	20	NA	NA	NA	28.51317	2.716233	11.81089	0
PRI	2014	21	NA	NA	NA	28.98146	2.284018	11.88102	0
PRI	2015	20	NA	NA	NA	29.76349	1.543381	11.95091	0
PRI	2016	20	NA	NA	NA	30.62716	2.540155	12.02063	0
PRI	2017	21	NA	NA	NA	31.10876	2.453635	12.09022	0
PRI	2018	NA	NA	NA	NA	31.61507	2.453551	12.15973	0
PRI	2019	NA	NA	NA	NA	32.91687	2.453467	12.22921	0

Country (ISO code)	Year	Maternal mortality	Infant mortality	Neonatal mortality	Under-5 mortality	GDP per capita	Rainfall	Male education	Armed conflict
CIV	2000	704	95.6	45.8	143.3	0.9867773	NA	NA	0
CIV	2001	706	93.7	45.2	140.1	0.9747800	NA	NA	1
CIV	2002	713	91.6	44.6	136.7	1.0209505	NA	NA	0
CIV	2003	711	89.6	44.0	133.1	1.1730639	NA	NA	1
CIV	2004	709	87.5	43.4	129.5	1.2677649	NA	NA	1
CIV	2005	704	85.2	42.7	125.6	1.2670872	NA	NA	1
CIV	2006	699	83.4	42.0	122.0	1.3035650	NA	NA	1
CIV	2007	698	81.1	41.3	118.2	1.4512326	NA	NA	0
CIV	2008	694	78.7	40.6	114.2	1.6833377	NA	NA	0
CIV	2009	700	76.6	40.0	109.9	1.6388048	NA	NA	0
CIV	2010	701	74.7	39.3	106.3	1.6541778	NA	NA	0
CIV	2011	703	72.9	38.7	103.1	1.7017047	NA	NA	1
CIV	2012	702	71.0	38.0	100.1	1.6493017	NA	NA	1
CIV	2013	691	69.1	37.4	97.0	1.9030543	NA	NA	0
CIV	2014	676	67.3	36.8	93.6	2.1240196	NA	NA	0
CIV	2015	658	65.7	36.2	90.8	1.9415663	NA	NA	0
CIV	2016	636	64.4	35.6	88.4	1.9808781	NA	NA	0
CIV	2017	617	62.9	35.0	86.0	2.0761480	NA	NA	1
CIV	2018	NA	61.3	34.5	83.2	2.2754959	NA	NA	0
CIV	2019	NA	59.4	33.9	80.3	2.2388110	NA	NA	0

As shown in @missingness-by-country, some countries may have missing data in multiple variables. For example Puerto Rico with mortality outcomes data, Ivory Coast in weather and education variables.

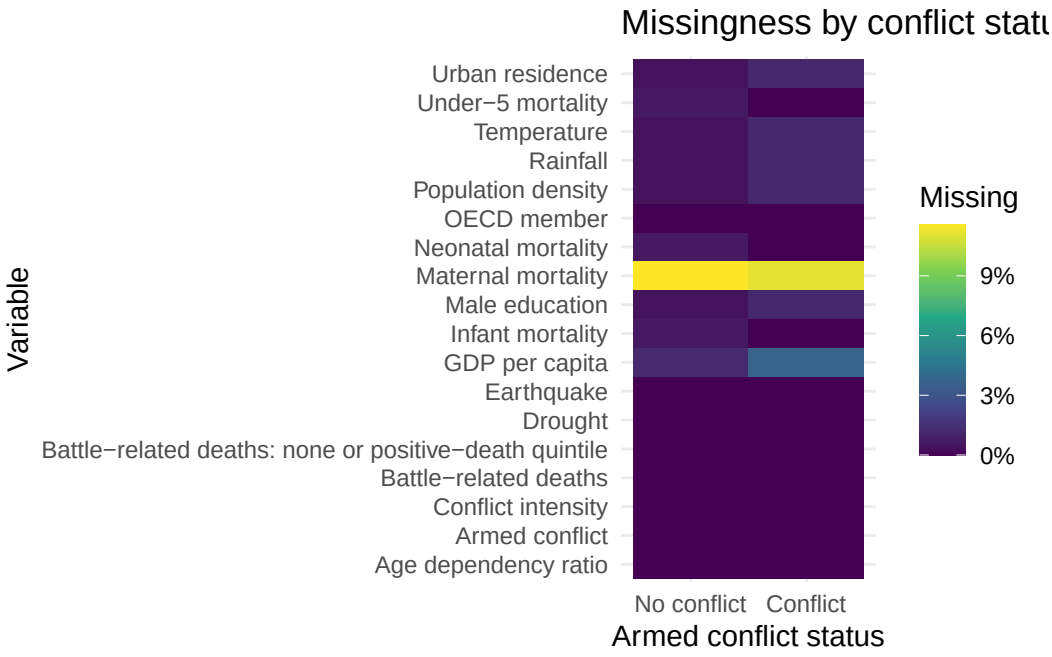


Figure 4: Proportion missing for each variable by conflict status.

conflict	any_missingness	n	proportion_missing
0	FALSE	2608	0.8647215
0	TRUE	408	0.1352785
1	FALSE	597	0.8480114
1	TRUE	107	0.1519886

Missingness is slightly higher in country-years with armed conflict. There are 3205 observations with no missingness in any variables.

Relationship between conflict and mortality outcomes

This descriptive plot compares mortality outcomes across the observed conflict indicator. It is not an adjusted or causal comparison.

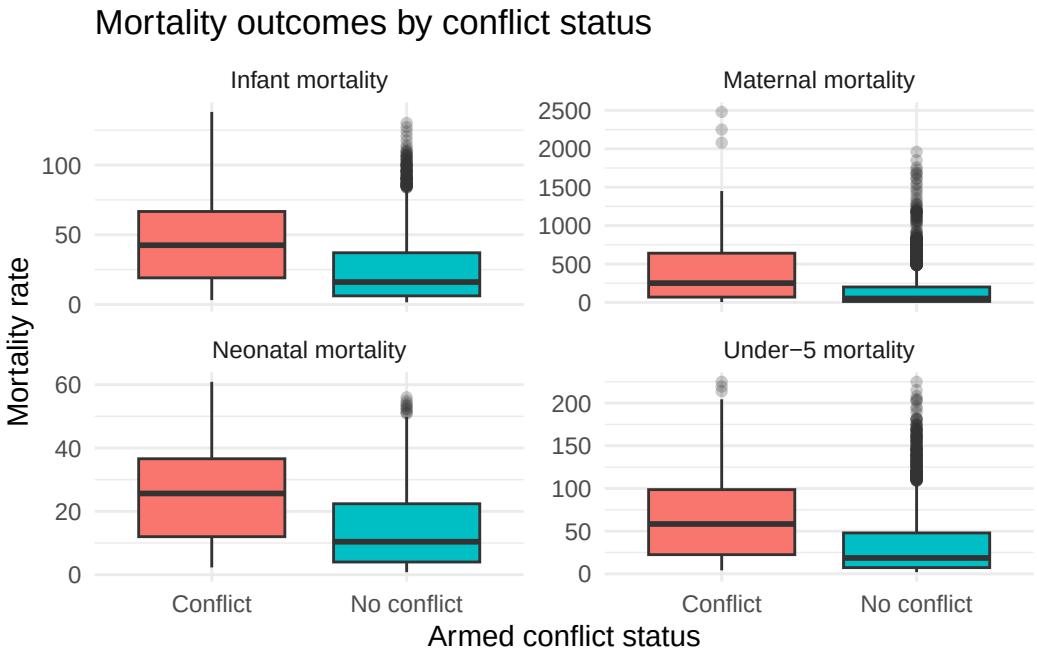


Figure 5: Mortality outcomes by conflict status.

Descriptive summary table of all variables in dataset by binary armed conflict status

- all continuous variables summarized by median and IQR

Descriptive summary table of all variables in dataset by binary armed conflict status

- all continuous variables summarized by mean and sd

Variable	Total N	Missing N	Level or summary	No conflict	Armed conflict	Total
Total N (%)				3016 (81.1)	704 (18.9)	3720
Maternal mortality	3294 (88.5)	426	Median (IQR)	51.0 (12.0 to 201.0)	251.5 (69.0 to 641.0)	66.0 (17.0 to 299.8)
Infant mortality	3700 (99.5)	20	Median (IQR)	16.1 (6.2 to 37.1)	42.5 (19.1 to 66.7)	18.9 (7.6 to 44.5)
Neonatal mortality	3700 (99.5)	20	Median (IQR)	10.4 (4.0 to 22.4)	25.6 (12.0 to 36.6)	12.1 (4.9 to 25.3)
Under-5 mortality	3700 (99.5)	20	Median (IQR)	18.6 (7.3 to 47.9)	58.4 (22.5 to 98.7)	22.2 (9.0 to 61.3)
GDP per capita	3658 (98.3)	62	Median (IQR)	5.1 (1.6 to 17.3)	1.3 (0.6 to 4.0)	4.1 (1.2 to 13.2)
OECD member	3720 (100.0)	0	0	2420 (80.2)	664 (94.3)	3084 (82.9)
			1	596 (19.8)	40 (5.7)	636 (17.1)
			(Missing)	0	0	0
Population density	3700 (99.5)	20	Median (IQR)	29.3 (14.8 to 41.8)	23.3 (14.5 to 36.1)	27.5 (14.8 to 40.7)
Urban residence	3700 (99.5)	20	Median (IQR)	30.4 (16.3 to 42.0)	29.4 (19.9 to 39.7)	30.3 (17.3 to 41.7)
Age dependency ratio	3720 (100.0)	0	Median (IQR)	53.9 (47.3 to 71.3)	74.5 (53.0 to 90.6)	55.5 (47.9 to 77.1)
Male education	3700 (99.5)	20	Median (IQR)	8.8 (6.5 to 11.2)	6.6 (4.6 to 8.3)	8.4 (5.9 to 10.8)
Temperature	3700 (99.5)	20	Median (IQR)	21.5 (11.6 to 25.7)	23.6 (19.8 to 26.5)	22.0 (12.9 to 25.9)
Rainfall	3700 (99.5)	20	Median (IQR)	1.0 (0.6 to 1.7)	1.0 (0.4 to 1.5)	1.0 (0.6 to 1.7)
Drought	3720 (100.0)	0	0	2783 (92.3)	612 (86.9)	3395 (91.3)
			1	233 (7.7)	92 (13.1)	325 (8.7)
			(Missing)	0	0	0
Earthquake	3720 (100.0)	0	0	2821 (93.5)	589 (83.7)	3410 (91.7)
			1	195 (6.5)	115 (16.3)	310 (8.3)
			(Missing)	0	0	0
Battle-related deaths	3720 (100.0)	0	Median (IQR)	0.0 (0.0 to 0.0)	386.5 (86.8 to 1446.0)	0.0 (0.0 to 2.0)
Conflict intensity	3720 (100.0)	0	No conflict (<25 deaths)	3016 (100.0)	0 (0.0)	3016 (81.1)
			Minor conflict (25-999 deaths)	0 (0.0)	479 (68.0)	479 (12.9)
			War (>=1,000 deaths)	0 (0.0)	225 (32.0)	225 (6.0)
			(Missing)	0	0	0
Battle-related deaths: none or positive-death quintile	3720 (100.0)	0	None (0 deaths)	2730 (90.5)	0 (0.0)	2730 (73.4)
			Positive deaths: quintile 1	198 (6.6)	0 (0.0)	198 (5.3)
			Positive deaths: quintile 2	88 (2.9)	110 (15.6)	198 (5.3)
			Positive deaths: quintile 3	0 (0.0)	198 (28.1)	198 (5.3)
			Positive deaths: quintile 4	0 (0.0)	198 (28.1)	198 (5.3)
			Positive deaths: quintile 5	0 (0.0)	198 (28.1)	198 (5.3)
			(Missing)	0	0	0

Variable	Total N	Missing N	Level or summary	No conflict	Armed conflict	Total
Total N (%)				3016 (81.1)	704 (18.9)	3720
Maternal mortality	3294 (88.5)	426	Mean (SD)	166.9 (258.4)	396.8 (398.1)	210.6 (303.8)
Infant mortality	3700 (99.5)	20	Mean (SD)	24.9 (23.9)	46.0 (29.7)	28.9 (26.4)
Neonatal mortality	3700 (99.5)	20	Mean (SD)	14.0 (11.8)	25.2 (13.9)	16.2 (13.0)
Under-5 mortality	3700 (99.5)	20	Mean (SD)	34.3 (38.0)	66.9 (49.5)	40.5 (42.4)
GDP per capita	3658 (98.3)	62	Mean (SD)	13.4 (18.6)	3.2 (5.0)	11.5 (17.4)
Population density	3700 (99.5)	20	Mean (SD)	31.2 (21.1)	27.8 (19.0)	30.6 (20.8)
Urban residence	3700 (99.5)	20	Mean (SD)	30.8 (18.4)	30.4 (13.8)	30.7 (17.6)
Age dependency ratio	3720 (100.0)	0	Mean (SD)	59.4 (17.3)	72.6 (21.2)	61.9 (18.9)
Male education	3700 (99.5)	20	Mean (SD)	8.7 (3.0)	6.5 (2.6)	8.3 (3.0)
Temperature	3700 (99.5)	20	Mean (SD)	19.0 (7.6)	22.2 (5.5)	19.6 (7.3)
Rainfall	3700 (99.5)	20	Mean (SD)	1.2 (0.8)	1.1 (0.7)	1.2 (0.8)
Battle-related deaths	3720 (100.0)	0	Mean (SD)	0.7 (3.0)	1905.1 (6489.7)	361.1 (2918.5)